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Research Paper

Impact of using a heat transfer fluid pipe in a metal hydride-phase change material tank

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HIGHLIGHTS

- The feasibility study of employing a HTF pipe in a MH tank that integrates a PCM is made.
- A 3D mathematical model is developed and is used for simulating various configurations of MH tank.
- The influence of HTF type and its mass flow rate on hydrogen filling time is evaluated.

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ABSTRACT

This study evaluates numerically the feasibility of employing a Heat Transfer Fluid (HTF) pipe in a Metal Hydride (MH) tank that integrates a Phase Change Material (PCM) heat exchanger. A 3D mathematical model is developed for this system and is used for simulating various configurations (Case 1: MH tank with PCM heat exchanger, Case 2: MH tank with PCM heat exchanger and open HTF pipe and Case 3: MH tank with PCM heat exchanger and closed HTF pipe). For each case, advantages and limitations are evaluated with respect to their heat transfer performance and filling time of hydrogen vis-a-vis heat storage capacity. In addition, the effects of the HTF type and its mass flow rate on the performance of the MH tank are studied. Besides showing a thermal coupling between MH bed, HTF pipe and PCM medium, the computational results indicate that the use of HTF pipe in MH-PCM tank is a tradeoff between reducing the hydrogen filling time and compromising the heat storage capacity. The results indicate that compared to the Case 1 (without HTF pipe) the filling time can be reduced by 94% for the Case 2 where the HTF pipe extracts 70% of the heat of reaction and delivers it to environment as waste heat. However, the filling time is reduced by only 72% for the Case 3 where all the heat of reaction is stored in PCM.

1. Introduction

Taking the current energy crisis and the environmental issues into consideration, recent technologies for efficient utilization of renewable energy sources have received much research interest. One of the major issues that need to be addressed in using renewable energy resources is that they are usually site-specific and intermittent. Because of the intermittent nature of the renewable energy sources, hydrogen has been used as a secondary energy carrier due to its abundance and environmental friendliness. Since hydrogen is a promising energetic vector, massive stationary hydrogen storage systems have been using this source [1]. Recent studies have focused on hydrogen storage systems in order to attain sufficient efficiency in hydrogen storage [2]. Hydrogen stored in Metal Hydride (MH) form is a potential candidate compared to other conventional methods of hydrogen storage such as compressed gas or liquid form through pressurization or liquefaction, which are expensive and energy intensive processes with many significant safety issues [3]. Further, MHs are considered as a frugal and viable solution for storing hydrogen, because of its reliability and high volumetric energy density compared to that of the conventional methods. Among the metals and alloys used as MHs, pure magnesium is considered a superior choice for its hydrogen storage capacity; which is capable to store 7.6 wt% of hydrogen [4]. An Mg based alloy, Mg_2Ni, which is capable to store 3.6 wt% of hydrogen is another example of MHs suitable for large-scale storage units. The main drawback of these hydrogen storage

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Nomenclature

A_a	absorption coefficient
B_a	absorption coefficient
C_a	material constant, s^{-1}
C_p	specific heat, $J\ kg^{-1}\ K^{-1}$
E_a	activation energy, $J\ mol^{-1}$
F	liquid fraction
H	enthalpy, $J\ kg^{-1}$
m	mass, kg
M	molecular weight, $g\ mol^{-1}$
$\vec{n}$	normal vector
P	pressure, Pa
R	universal gas constant, $J\ mol^{-1}\ K^{-1}$
t	time, s
T	temperature, K
V	volume, m^3
$\vec{V}$	velocity of HTF, ms^{-1}
X	absorbed hydrogen amount, w%
ΔH	reaction enthalpy, $J\ kg^{-1}$
ΔS	reaction entropy, $J\ K^{-1}\ kg^{-1}$

Greek symbols

ε	porosity
λ	thermal conductivity, $W\ m^{-1}\ K^{-1}$
μ	dynamic viscosity, Pa s
ρ	density, $kg\ m^{-3}$

Subscripts

0	initial
a	absorption
eff	effective
eq	equilibrium
HTF	heat transfer fluid
liq	liquidus
m	melting
MH	metal hydride
ref	reference
sol	solidus

materials is a requirement of high-temperature heat flux, necessitating an external source of heat. To supply the heat needed to initiate sorption process, MH tank should be coupled with another systems such as high temperature fuel cells [5,6], thermal power plants or external heat source [7] which release a part of thermal energy as waste. If the power generation systems which release waste heat are not available, efficient ways are needed to supply the heat. Since the absorption process is exothermic and desorption process is endothermic reactions, heat of reaction during absorption process may be stored and further used in desorption process. Use of Phase Change Material (PCM) which can store latent heat is a recent and an attractive solution to resolve this issue [8–10,4]. Recent literature review reveals only a limited work on the application of Mg metal hydride in heat storage systems; furthermore, these studies were made only on a few simple configurations, where MH bed is in direct-contact with a PCM heat exchanger [8,9]. Some modeling tools for the prediction of the evolution of the various physical parameters in a MH-PCM system was presented by Marty et al. [10]. A fair agreement of numerical data with experimental data was found in their study [4]. In a previous work [8], present authors have studied an MgH_2 system to check the possibility to store the heat of reaction in charging process by using a PCM in order to reuse heat of reaction in discharging process. Recently, Ben Mâad et al. [9] have studied numerical simulation of sorption cycles for low temperature MH using PCM for heat recovery. Experimental and simulation studies have concluded low thermal conductivity of MHs and PCMs as a main drawback, since it reduces the heat transfer within the system. Thus, for practical applications, heat transfer enhancement is a vital issue for designing large scale MH-PCM tanks to store hydrogen and heat of reaction. Many enhancements techniques have been suggested to improve the heat transfer within MH beds or PCM units. These heat transfer enhancements methods are classified based on two approaches. The first approach involves augmenting the overall thermal conductivity of medium (PCM or MH bed) either by placing a copper wire-net matrix [11], compressing copper-encapsulated powders [12], combining materials with natural graphite [13] or integrating aluminum foam [14]. The second approach is to optimize heat exchanger design incorporated within the MH tank either by using plate-fin exchanger [15], concentric inner finned tubes [16,17], helical-coil heat exchanger with fins [18–20] or a heat pipe [21–24]. However, most of these studies are mainly focused on the heat transfer within MH bed, while the heat transfer within the internal HTF pipe or the external cooling jacket was paid little attention. For simplicity, these numerical simulations often assumed: (i) negligible temperature change over the diameter of the heat exchange pipe, so that the governing equation is simplified to one dimension [23,24] and (ii) a constant convective heat transfer coefficient along the inner surface of the heat pipe, in modeling heat transfer in HTF pipes [23,24].

Moreover, most of these numerical studies on MH tanks use a HTF to extract the reaction heat and deliver it in environment as waste heat [16–19,23–25]. However, the dissipated heat must be supplied again to the MH bed during discharge process. Therefore, an external heat source is needed to supply the heat of desorption. However, this reduces the global efficiency of the MH tanks in terms of thermal energy. A MH tank that integrates a PCM - as a heat storage unit which replaces the usage of an external heat source - is economical for large scale hydrogen storage systems. One major issue that needs to be addressed in these systems is that most of the MHs and the PCMs have low thermal conductivity, demanding an appropriate heat transfer enhancement solution. So, the idea is to propose a way to quickly transport the heat released during the charging process to a PCM via a HTF and to recover it thereafter. Despite many studies that have been conducted and reported on thermal couplings, none of the studies contain three domains such as MH bed, PCM medium and a HTF pipe. Also, currently there is no information on the effect of integrating HTF pipe in the MH-PCM unit and the effect of employing HTF for transferring the heat to/from the PCM on the hydrogen storage performance. Thus, the objective of this study is to compare three different configurations; Case 1: MH bed in a direct-contact with PCM heat exchanger, Case 2: MH bed in a direct-contact with PCM heat exchanger and with HTF pipe to accelerate heat transfer and Case 3: MH-PCM tank with HTF pipe for heat storage and recovery to/from PCM. Hence, the aim is to demonstrate the effects of using HTF pipe to improve the thermal performance of hydrogen and heat storage tank. For each of these cases, Mg_2Ni alloy is used as a hydride bed and the sodium nitrate ($NaNO_3$) is selected as the heat storage medium (PCM).

In this article three design configurations of the MH-PCM system are described at first. Then, a 3D mathematical model of heat and mass transfer within the MH bed, the PCM domain and the HTF pipe is detailed. The computational results for analyzing the MH-PCM systems performance focusing on heat transfer enhancement and heat storage performance are discussed further. Finally, the effects of the HTF type and its mass flow rate on the performance of the MH tank are discussed.

2. Different configurations of the MH-PCM system

In the present study, three different configurations of an insulated MH-PCM system are investigated:

Case 1: MH powder is filled inside a closed enclosure (cylindrical tank) as shown in Fig. 1a. The cylinder of MH bed is surrounded by a cylindrical jacket filled with PCM to store the heat of reaction. In this case, a direct-contact between the MH bed and the PCM is made in such a way that, total heat of reaction can be stored in PCM. A central filter tube is used to introduce or recover hydrogen (Fig. 1a). The whole tank is insulated to prevent heat loss through the outer wall of the cylindrical jacket.

Case 2: The configuration of the MH-PCM tank is similar to that in Case 1, but twelve heat exchanger tubes are placed in a circular array around the hydrogen filter tube inside the MH bed. Heat

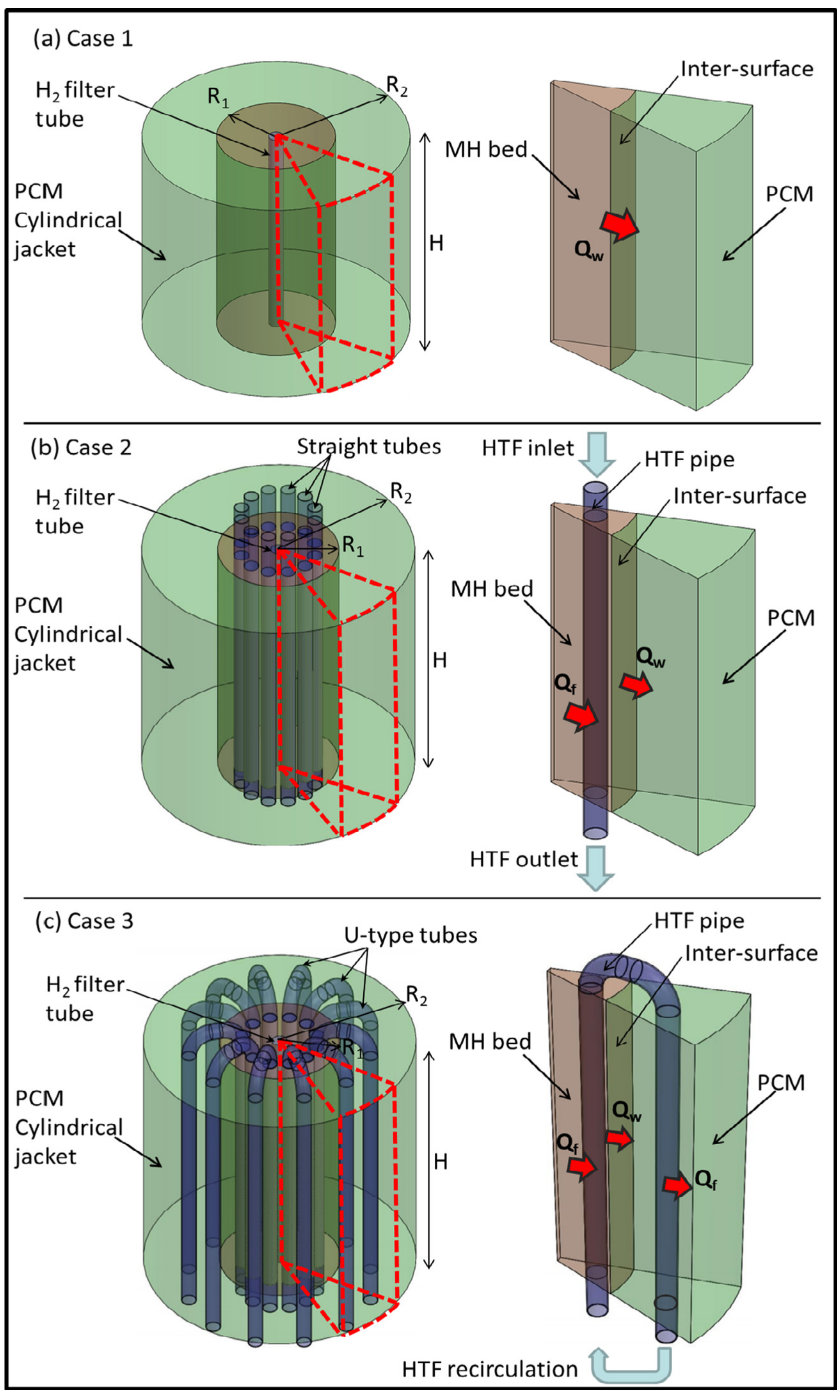

Fig. 1. Geometries of various configurations of MH-PCM tank: (a) without HTF pipe (Case 1), (b) with open HTF pipe (Case 2) and (c) with closed HTF pipe (Case 3).

Table 1
Geometric parameters of the simulated configurations.

Parameter	Value (mm)		
	Case 1	Case 2	Case 3
R_1	36	40	40
R_2	83.4	85.1	86.9
H	164.3	164.3	164.3
Radius of filter tube	6	6	6
Radius of HTF pipe	–	5	5

transfer fluid (HTF) flows through these tubes to extract the heat of reaction and delivers it to environment as waste heat. During discharge process the heat is supplied again to the MH bed by the HTF. The schematic of this multi-tubular tank is shown in Fig. 1b. In this case, heat of reaction released in the proximity of the heat exchanger tubes is extracted from the MH bed by the HTF while the heat released near the external wall of MH bed is stored in the PCM cylindrical jacket.

Case 3: The configuration of the MH-PCM tank is similar to that in Case 1, but it is equipped with a circular array of twelve U-type heat exchanger tubes (Fig. 1c). The first half of each U-type tube is inserted through the MH bed and the second half is through the PCM domain. The HTF flows through these U-type tubes extract the heat of reaction from MH bed and transfer it to the PCM. In this case, heat of reaction is totally extracted from the MH bed by the HTF tubes and the PCM in direct contact with MH bed.

All the geometric parameters for these configurations are presented in Table 1.

3. Mathematical formulation

For this study, the simulation is simplified by considering the axial symmetry of MH-PCM systems of the computational domain [17,29]. The computational domain (a sector of 1/12th) used for a 3D simulation of the MH-PCM tank is shown in Fig. 1. The main assumptions are similar to those used in previous numerical studies done on the magnesium based MH reactors [17,25–29], which are:

i. The local thermal equilibrium is valid [25,27,29].
ii. Thermo-physical properties are constant [27,29].
iii. The thermal resistance and the thermal capacity of the walls of MH-PCM system are neglected compared to those of the MH bed [8,27,29].
iv. Hydrogen flow is assumed negligible during the reaction process. A dimensionless number N was developed by Chaise et. al. [26] to evaluate the influence of the hydrogen flow on the reaction process. It was defined that when N is smaller than 0.01, the hydrogen flow can be neglected without introducing significant error to the simulation results.
v. Radiative heat transfer is neglected [8,25–29].
vi. The liquid PCM is highly viscous; hence the convection is negligible [8,29,30].

3.1. Governing equations for the Mg_2Ni hydride

3.1.1. Energy equation

Assuming thermal equilibrium between hydrogen and Mg_2Ni alloy, the energy equation is:

$$\frac{\partial((\rho \cdot C_P)_{eff,MH} \cdot T)}{\partial t} = \nabla \cdot (\lambda_{eff,MH} \nabla(T)) + S \tag{1}$$

where the coefficient $(\rho \cdot C_P)_{eff,MH}$ is given by the following expression [27,29]:

$$(\rho \cdot C_p)_{eff,MH} = \varepsilon_{MH} \cdot \rho_{H_2} \cdot C_{P_{H_2}} + (1 - \varepsilon_{MH}) \cdot \rho_{MH} \cdot C_{P_{MH}} \tag{2}$$

where ε_{MH} is the porosity of MH bed, which is assumed constant ($\varepsilon = 0.5$) and uniform over the MH bed [34].

The effective thermal conductivity of the MH bed $\lambda_{eff,MH}$ is given by [8,27,29]:

$$\lambda_{eff,MH} = (1 - \varepsilon_{MH}).\lambda_{MH} + \varepsilon_{MH} \cdot \lambda_{H_2} \tag{3}$$

The source term S at the right-hand side of Eq. (1), which is proportional to the reaction rate, is given by:

$$S = \frac{\rho_{MH}.(1 - \varepsilon_{MH})}{M_{H_2}} \frac{dX}{dt} \Delta H_{MH} \tag{4}$$

where X expressed in *wt%* is the absorbed hydrogen amount, dX/dt is the kinetic reaction and ΔH_{MH} is enthalpy of reaction.

3.1.2. Kinetic reaction

Thermodynamic and kinetic aspects are treated by calculating $P_{a,eq}$, dX/dt and X in each cell. The kinetic equation for the absorption is given as follows a [27]:

$$\frac{dX}{dt} = C_a \exp\left(-\frac{E_a}{R.T}\right)\left(\frac{P_{H_2} - P_{a,eq}}{P_{a,eq}}\right)(X_{\max} - X) \tag{5}$$

where R is the universal gas constant, T the bed temperature, and P_{H_2} is the hydrogen pressure, $P_{a,eq}$ is the equilibrium pressure for absorption and X_{max} expressed in *wt%* is the maximum absorbable hydrogen amount.

The equilibrium pressure $P_{a,eq}$ in Eq. (5) is determined by the Van't Hoff equation and is considered to be a function of temperature as shown:

$$\frac{P_{a,eq}}{P_{ref}} = 10^{-5} \exp\left(A_a - \frac{B_a}{T}\right) \tag{6}$$

where the reference pressure $P_{ref} = 1bar$, and A and B are plateau pressure coefficients summarized in Table 2.

3.2. Governing equations for the PCM

The sodium nitrate ($NaNO_3$) is selected as the heat storage medium (PCM). Some details of the selection procedure of the PCM are given in the previous paper [29]. In a phase change process, the energy equation based on the enthalpy formulation can be expressed in terms of total volumetric enthalpy and temperature for constant thermo-physical properties, as follows [8,29].

$$\frac{\partial H}{\partial t} = \nabla \cdot (\lambda_{PCM} \nabla(T)) \tag{7}$$

where H is the total volumetric enthalpy of the PCM and λ_{PCM} is the effective thermal conductivity of the PCM.

Table 2
Physical proprieties of the Mg_2Ni hydride and the PCM ($NaNo_3$) [27,29].

Parameters	Mg_2Ni hydride	PCM ($NaNo_3$)
Thermal conductivity, λ ($W\,m^{-1}\,K^{-1}$)	1.33	0.48
Specific heat, C_p ($J\,kg^{-1}\,K^{-1}$)	1414	1820
Density, ρ ($kg\,m^{-3}$)	3200	2260
Porosity, ε	0.5	–
Specific enthalpy of melting, ΔH ($J\,g^{-1}$)	–	174
Melting temperature, T_m (K)	–	580
Solidus temperature, T_{sol} (K)	–	579
Enthalpy of reaction, ΔH ($J\,mol^{-1}\,H_2$)	−64,000	–
Entropy of reaction, ΔS ($J\,K^{-1}\,mol^{-1}\,H_2$)	−122	–
Absorption plateau pressure coefficient A_a	26.481	–
Absorption plateau pressure coefficient B_a	7552.5 K	–
Absorption rate constant C_a	175.31 s^{-1}	–
Activation energy for absorption, E_a (J)	52.205 k j mol^{-1}	–

$$H(T) = h(T) + \rho_{PCM} \cdot \Delta H_{PCM} \cdot F(T) \tag{8}$$

where F is the liquid fraction, ΔH_{PCM} is the latent heat of the PCM, and h can further be written as [29]:

$$h(T) = \int_{T_m}^{T} \rho_{PCM} \cdot C_{P_{PCM}} dT \tag{9}$$

The liquid fraction can be defined as follows [28]:

$$F = \begin{cases} 0 & \text{if} \quad T \leqslant T_{sol} \\ 1 & \text{if} \quad T \geqslant T_m \\ \frac{(T-T_{sol})}{(T_m-T_{sol})} & \text{if} \quad T_{sol} < T < T_m \end{cases} \tag{10}$$

where T_m is the melting temperature and T_{sol} is the solidus temperature.

Assuming all the heat of reaction can be stored with the PCM, the volume and the mass of PCM per unit kg of Mg_2Ni are expressed as follows [29]:

$$V_{PCM} = -\frac{wt\Delta H_{MH}}{\rho_{PCM} M_{H_2} \Delta H_{PCM}} \tag{11}$$

$$m_{PCM} = -\frac{wt\Delta H_{MH}}{M_{H_2} \Delta H_{PCM}} \tag{12}$$

where ρ is the density, ΔH is the enthalpy and M is the molar mass of hydrogen.

3.3. Governing equations for HTF

The energy equation of the HTF is as follows:

$$\rho_{HTF} C_{P\ HTF} \frac{\partial \cdot T_{HTF}}{\partial t} + \nabla \cdot (\rho_{HTF} \vec{V}_{HTF} \cdot T_{HTF}) = \nabla \cdot (\lambda_{HTF} \nabla T_{HTF}) \tag{13}$$

where $\vec{V}_{HTF}$ is the velocity of the HTF.

The flow of the HTF can be described by the Navier-Stokes equation:

$$\rho_{HTF} \frac{\partial \vec{V}_{HTF}}{\partial t} + \rho_{HTF} \overline{\overline{\nabla}} \vec{V}_{HTF} \cdot \vec{V}_{HTF} = \nabla \cdot [\mu_{HTF} (\nabla \vec{V}_{HTF} + (\nabla \vec{V}_{HTF})^T)] - \nabla p_{HTF} \tag{14}$$

3.4. Initial and boundary conditions

3.4.1. Initial conditions

During charging, the MH temperature and the hydrogen pressure as well as the PCM temperature are assumed to be uniform. Therefore,

$$T_0 = 579\ \text{K},\ P_0 = 15\ \text{bar} \tag{15}$$

3.4.2. Boundary conditions

The boundary conditions are defined as follows:

(1) Adiabatic wall (or symmetric boundary condition):

$$\nabla T_{MH} \cdot \vec{n} = \nabla T_{PCM} \cdot \vec{n} = 0 \tag{16}$$

where $\vec{n}$ represents the normal vector to the corresponding wall.

(2) Hydrogen inlet boundary condition:

$$\nabla T \cdot \vec{n} = 0 \tag{17}$$

(3) The inter-phase boundary conditions for the domain MH bed/PCM:

$$\lambda_{MH} \cdot \nabla T_{MH} \cdot \vec{n} = \lambda_{PCM} \cdot \nabla T_{PCM} \cdot \vec{n} \tag{18}$$

(4) At the interface between HTF and PCM or MH bed, no-slip condition is specified for velocities and coupled temperature to allow conjugate heat transfer.

(5) Inlet/Outlet boundary conditions of the HTF depend on the configurations, which are defined as follows:

Case 2: -At the inlet, mass flow rate and temperature of HTF are specified as 0.005 kg/s and 579 K respectively.

-At the outlet, the mass flow rate is assumed to be delivered at an atmospheric pressure and the gradient of the HTF temperature is zero.

Case 3: HTF is circulated out from PCM domain (outlet) and it is circulated back to the MH bed (inlet) with the same mass flow rate (0.005 kg/s) and temperature as shown in Fig. 1c. Recirculation boundary condition is specified at the inlet.

3.5. Model validation and computational procedure

The mathematical model developed in this study was validated through a similar model used in a previous study [29] which comprises of two components, MH bed and PCM storage, and allows a conjugate heat transfer between MH bed and PCM.

The computational domains were created in the commercial software GAMBIT 2.3.16. It was also used for meshing and specifying computational domains with initial and boundary conditions. Fluent 6.3.26 was used for simulation. User-defined functions (UDF) were written in C language to calculate the kinetic reaction rate of the hydriding process and the equilibrium pressure of MH bed. The first-order upwind differencing scheme was used for solving the momentum and energy equations.

4. Results and discussion

4.1. Impact of the HTF pipe in the MH-PCM tank

Each of the configurations is studied to understand the influence of integrating a HTF pipe on the performance of the MH-PCM tank by evaluating filling time of hydrogen and the percentage of heat of reaction stored in PCM. In order to find the best performance of the MHT-PCM systems, the quantity of MH and PCM, the boundary and the initial conditions (15 bar, initial temperature of 579 K) were kept the same for each of the three configurations. The molten-salt (Sandia Mix) is used as the HTF.

4.1.1. Study on configuration-1

The absorption process is simulated for Case 1 in which the heat of reaction (heat flow Q_w) is transferred to the PCM medium directly through the inter-surface wall. The three dimensional temperature distributions in MH-PCM system and the liquid fraction in PCM domain at different instants (7000 s, 22,000 s and 50,000 s) are shown in Fig. 2. The temperature is quite uniform over the entire MH bed except at the inter-surface of the MH bed and the PCM domains because the heat released from the MH bed is transferred to PCM through the wall (Fig. 2a). Near the inter-surface, the MH bed temperature is lower compared to a region away from the PCM. The melting process of the PCM is faster in the vicinity of the inter-surface and leads to higher melting rate compared to region farther to the inter-surface (Fig. 2b). Hydriding MH bed requires large temperature gradient to conduct the heat of reaction to the PCM domain due to its low thermal conductivity. The high temperature prevailing farther from the PCM domain increases the corresponding equilibrium pressure, thereby reducing the reaction rate.

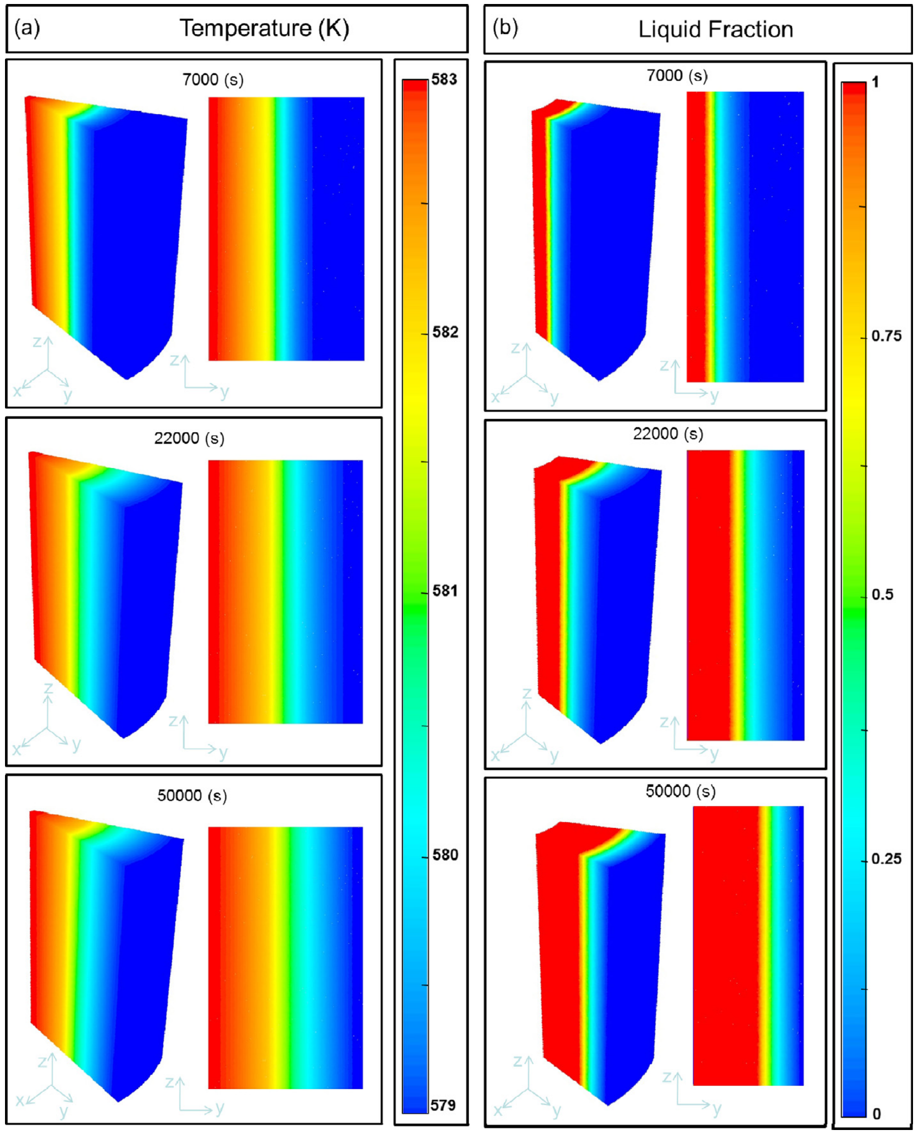

Fig. 2. Three dimensional distribution of (a) the temperature in the MH-PCM tank and (b) the liquid fraction in the PCM domain at 7000 s, 22,000 s and 50,000 s (for the Case 1).

For this configuration, Fig. 3 shows that the heat flow Q_w through the inter-surface rises at the beginning of the loading process and finally tends to zero. This behavior can be explained by the fact that the MH bed is completely hydrided. Thereafter, the generation of heat reaction is stopped.

4.1.2. Study on configuration-2

In Case 2, heat of reaction released in the proximity to the heat exchanger tube is extracted from the MH bed by HTF pipe and delivers it to the environment as waste heat (heat flow Q_f). While, the heat released near the lateral wall of MH bed is stored in the PCM cylindrical jacket (heat flow Q_w).

The three dimensional temperature distributions in MH-PCM system and the liquid fraction in PCM domain at different instants (1000 s, 2000 s and 2500 s) are shown in Fig. 4. It can be noted that the temperature at the vicinity of the HTF pipe is lower and further decreases gradually than that at the inter-surface of the PCM medium with the absorption process (Fig. 4a). This can be explained by the fact that the HTF flows through the tube extracts the reaction heat by the cooling effect of the fluid. On the other side at the inter-surface the heat is extracted also. Meanwhile, the melting front proceeds radially from the inter-surface of PCM-MH bed and extends to the external PCM domain (Fig. 4b).

The evolution of the heat flow (Q_w) transferred from MH bed to PCM medium and the heat flow (Q_f) delivered to environment as waste heat by HTF is shown in Fig. 5. It is observed that maximum percentage of reaction heat is extracted by HTF pipe and the rest of the heat is transferred and stored in PCM domain.

4.1.3. Study on configuration-3

During absorption process for the Case 3, all the heat of reaction within the MH bed is transferred to the PCM medium following two paths, one through the heat exchanger pipe where the HTF runs and the other directly through the inter-surface wall.

The three dimensional temperature distributions in MH-PCM system and the liquid fraction in PCM domain at different moments (7000 s, 12,000 s and 16,000 s) are shown in Fig. 6. Due to the exothermic effect of the hydrogen absorption process, the temperature of the MH bed increases rapidly. Moreover, it can be seen that the temperature of the MH bed near HTF pipe and at inter-surface is lower than that of other regions. It is observed from the temperature distribution in a cross-section of the simulated

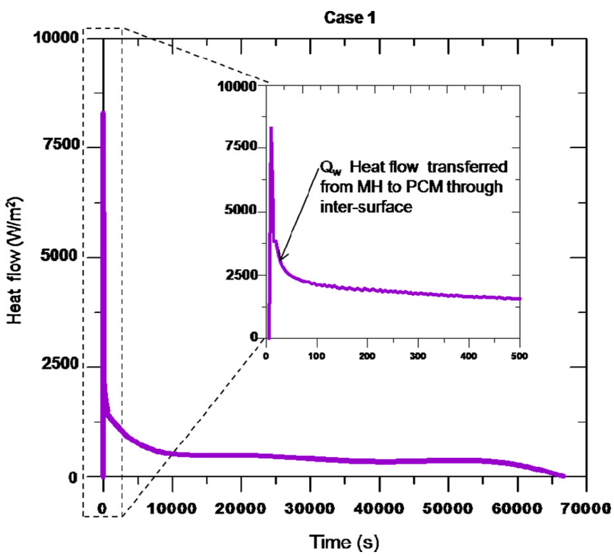

Fig. 3. Evolution of the heat flow for Case 1.

domain, the temperature of HTF increases along the flow direction in the MH bed and decreases in the PCM medium (Fig. 6a). This is due to the fact that the reaction heat is extracted at MH bed and it is delivered in PCM domain by HTF. Also, it is observed from the distribution of liquid fraction that the melting front initiated at the inter-surface is less fast than that at the vicinity of HTF pipe (Fig. 6b).

From Fig. 7 it can be noted that the maximum heat flow transferred from MH bed to PCM medium through the inter-surface wall (Q_w) presents 58% of all the heat reaction which is more than (Q_f) that through the HTF pipe (42%). Also, it is observed from Fig. 7 that after a substantial period of time, the heat flow released from the MH bed decreases and tends to zero since the MH bed saturates and thus the reaction rate decreases. In the PCM medium the same heat flow transferred from MH bed by HTF is extracted by the PCM and initiates melting process.

The evolution of the average temperatures of the HTF that flows through the pipe inside the MH bed and the PCM medium is shown in Fig. 8 (for Case 3). The mean temperature of the HTF increases sharply in the initial stage and reaches its maximum value. When the hydriding rate reaches the maximum, the maximum outlet

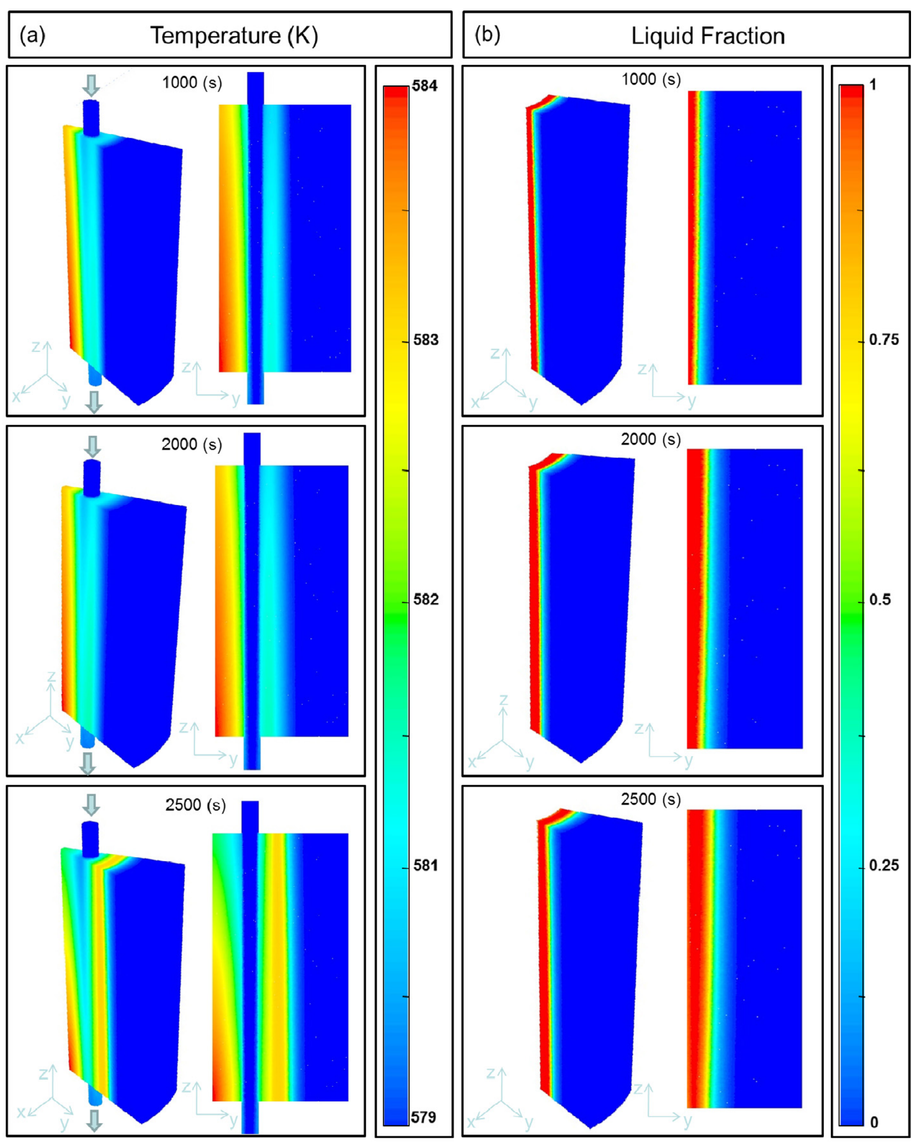

Fig. 4. Three dimensional distribution of (a) the temperature in the MH-PCM tank and (b) the liquid fraction in the PCM domain at 1000 s, 2000 s and 2500 s (for the Case 2).

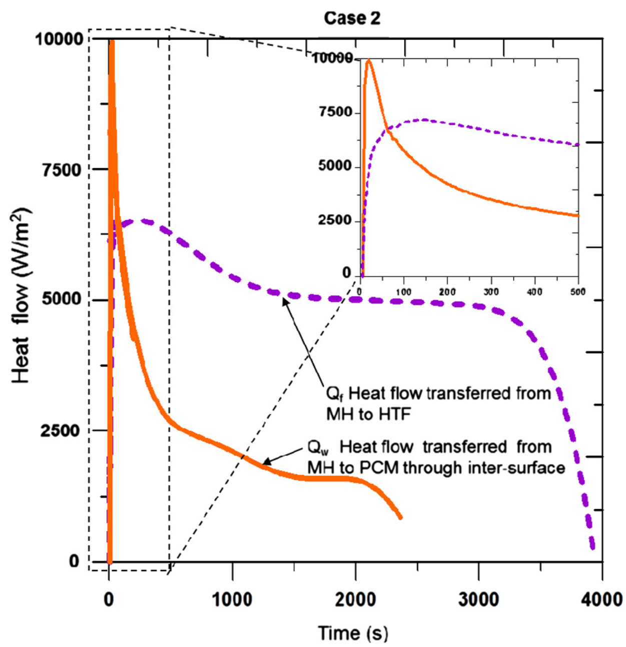

Fig. 5. Evolution of the heat flows for Case 2.

mean temperature values of the HTF will be 582.7 and 581.2 K, respectively. It can be noticed that the average temperature difference between HTF pipe in MH bed and HTF pipe in PCM medium is still remarkable ($\Delta T > 2$) when the absorption process is proceeds. Subsequently, the temperatures slowly decreases approaching the melting temperature of the PCM at the end of hydriding process. This result is attributed to the thermal coupling between the MH bed and the PCM domain, which leads to the storage of reaction heat by a PCM can significantly improve the energy efficiency of the MH hydrogen storage tank.

4.1.4. Comparison between different configurations

The performance of various configurations is compared to examine the benefits of employing a HTF pipe in a MH-PCM tank. The time evolution of the average hydrogen reacted fraction and the average liquid fraction of PCM for the three cases of MH-PCM tank is presented in Fig. 9.

Since, the hydriding process is controlled by the heat transfer rate from the MH bed to the PCM, it can be seen from Fig. 9a that the 90% of hydrogen saturation state is obtained at various durations for each configurations. The Case 1 follows a straight line trend and reaches 90% of saturation much later (53,270 s) whereas to reach the same state Case 2 and Case 3 requires only 3135 s and 14,685 s respectively. A comparison of the two novel cases (Case 2 and Case 3) with the basic configuration (Case 1) indicated 94% and 72% improvement in filling time respectively.

Also, Fig. 9 displays the comparison of the effects of integrating the HTF pipe inside the MH-PCM tank on the performance of the system. It can be clearly seen from Fig. 9a that almost at any moment the reacted fraction in Case 2 is higher than those in the Case 1 and 3. The hydriding process for the Case 2 is almost completed within 4000 s. In contrast, only 30% of the hydriding reaction for the Case 3 is completed at 4000 s. However, in Case 2 the almost of heat reaction (70%) is extracted by the HTF and delivered to environment as waste heat. Therefore this approach offers better heat transfer performance. Since, all the heat of reaction is not stored, thus the PCM is not melted completely and it is indicated by lowest melting fraction (0.3) at early stage of melting process. However, in Case 1 and Case 3 all the heat of reaction is extracted and stored in the PCM medium (Fig. 9b).

To summarize, the Case 2 (where a HTF flows through pipe to extract the heat of reaction and delivers it to environment as waste heat) can reduce the hydrogen filling time by 94% compared to the Case 1. However, Case 3 (where a HTF flows through U-type pipe extracts the heat of reaction from MH bed and store it in a PCM, after that it recirculates) increases the hydrogen filling time by 78% compared to the Case 2. Thus, filling time is more sensitive to the chosen approach. The selection of an appropriate configuration is the result of a tradeoff between desired filling time and the amount of the heat that can be stored. There are two alternatives which are either reducing the hydrogen filling time with decreasing in the energy performance of the system or increasing the energy performance with increasing the hydrogen filling time. The best policy depends on the desired application in terms of necessity where it is obligatory to foster any of these alternatives.

These results suggest that the novel approach in Case 3 is more desirable and economical for heat reaction storage in PCM during absorption process which replaces the usage of an external heat source during desorption process. However, further optimization may be needed for large scale hydrogen storage tanks. A complete optimization of the Case 3 should cover a few key parameters, including the thermal properties of the HTF and its mass flow rate. As a result, optimization parameters can decrease temperature difference between MH bed and HTF, or reduce thermal resistance of the system, which would improve the hydride reaction efficiency. Thus, the following sections will discuss this topic.

4.2. Effect of HTF type on hydride reaction efficiency

The selection of an appropriate HTF for the MH-PCM tank is a critical task and is usually done with respect to the operating temperature of the MH tank. HTF is one of the most important components for overall performance and efficiency of the MH-PCM tank. Since for large scale hydrogen storage a large amount of HTF is required to operate the MH-PCM tank, it is necessary to minimize the cost of HTF while maximizing its performance. Hence, investigation was performed to simulate the effect of HTF types using Case 3. The HTFs that used in this section are: (1) air, (2) water/steam, (3) thermal oil, and (4) molten-salt. The different physical properties, the cost and the rate of corrosion of all the possible HTFs in this topic are summarized in Table 3.

To directly compare the performances of these HTFs, the histograms in Fig. 10 shows a summary of the time interval for completing 90% of the hydrogen storage capacity, under identical conditions (HTF mass flow rate is 0.005 kg/s). Molten-salt as HTF clearly exhibited a faster rate (14,685 s) compared to that of other three HTF types, thermal oil (22,750 s), water/steam (27,390 s) and air (31,075 s). Thus, with the molten salt as HTF the hydrogen filling time was reduced by 35%, 46% and 52% over the cases with thermal oil, water/steam and air, respectively. As summarized in the Table 3 where is compared the various HTFs and consolidated properties such as viscosity, thermal conductivity, heat capacity. It is clear that air and Water/steam have very low dynamic viscosity and thermal conductivity compared to other HTFs [31]. The major disadvantages are the low heat capacity and heat transfer between these HTFs and the internal surfaces of the HTF pipe. To quickly transfer heat from or to the MH bed, the HTF must have the following characteristics: high thermal conductivity, high heat capacity for energy storage, low viscosity, low melting point, high boiling point and thermal stability [32].

The molten salt used in this study is known as 'Sandia Mix' ($NaNO_3$(9–18%)-KNO_3 (40–52%)-$LiNO_3$ (13–21%)-$Ca(NO_3)_2$ (20–27%)) [33]. It has relatively lower melting point (95 °C) and thermal stability limits higher than 500 °C. The viscosity of this

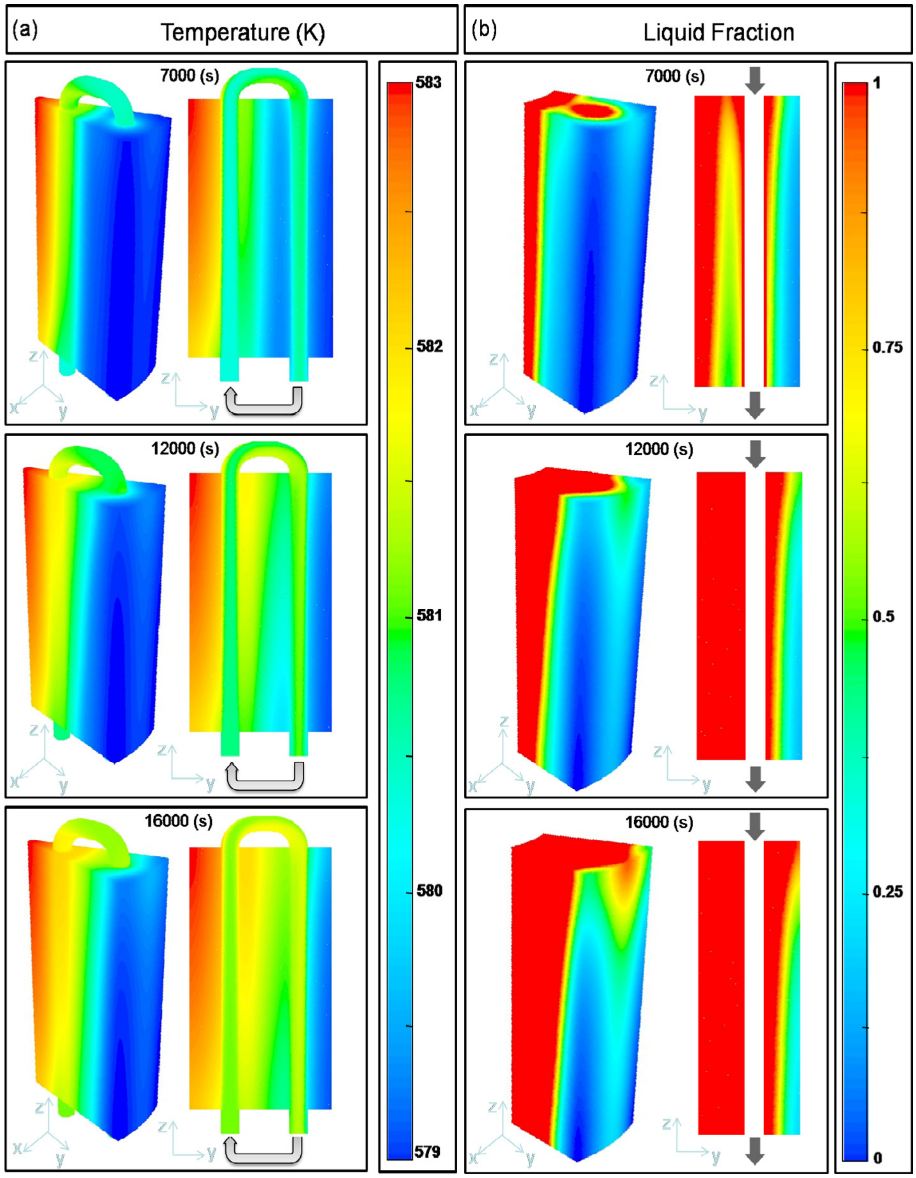

Fig. 6. Three dimensional distribution of (a) the temperature in the MH-PCM tank and (b) the liquid fraction in the PCM domain at 7000 s, 12,000 s and 16,000 s (for the Case 3).

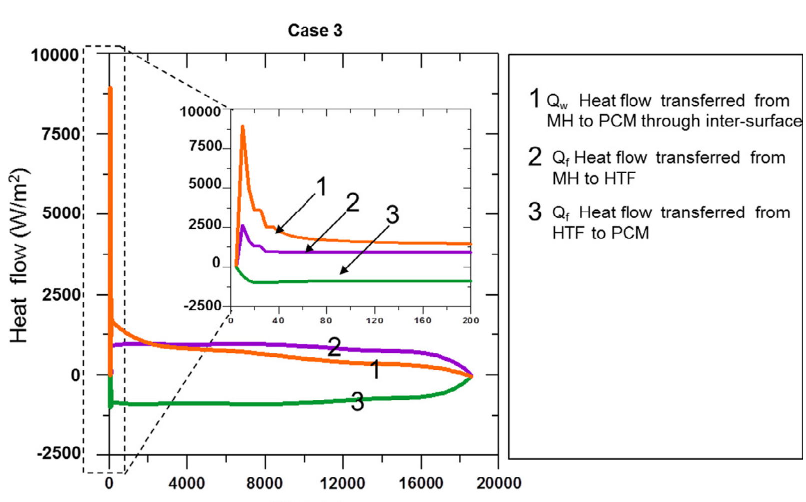

Fig. 7. Evolution of the heat flows for Case 3.

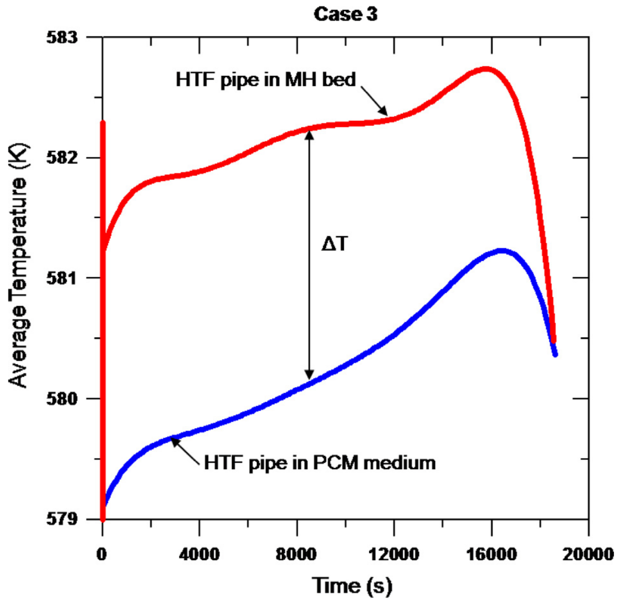

Fig. 8. Evolution of the average temperature in the HTF pipe for Case 3.

molten-salt is less than 0.003 Pa s, which is an acceptable value for this application. Even though the molten-salt is one of the very attractive HTF, its usage in MH tank is not studied well. The main drawbacks of this type of molten-salt that for the large scale applications it has the highest cost (0.62–0.81 \$/kg) compared to other HTFs. Also, molten-salt have a relatively high corrosive nature

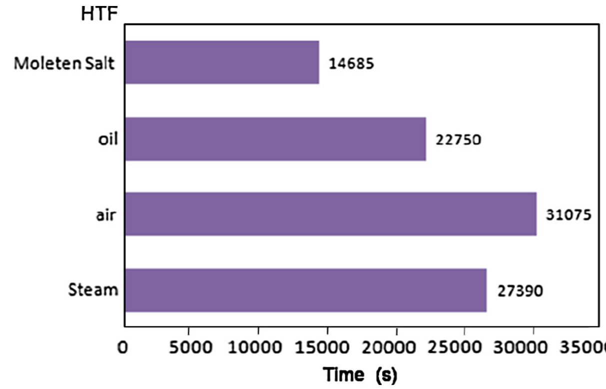

Fig. 10. Effect of HTF type on 90% hydrogen filling time (for Case 3).

because it is based on nitrates/nitrites. Despite of this molten-salt is one of the very attractive HTFs in viewpoint of better thermal properties. However, the safety of the tank and the risk of corrosion of HTF pipe limit the application of this HTF.

To summarize, molten-salt is being proposed as HTF in MH-PCM tanks for the best storage performance. However, the corrosion issues need to be resolved completely before the application is commercialized.

4.3. Effect of HTF mass flow rate on hydrogen storage performance

In the numerical simulation, the Case 3 is taken to evaluate the effect of HTF mass flow rate on hydrogen storage performance. All parameters used in earlier simulations, as shown in Table 2, were

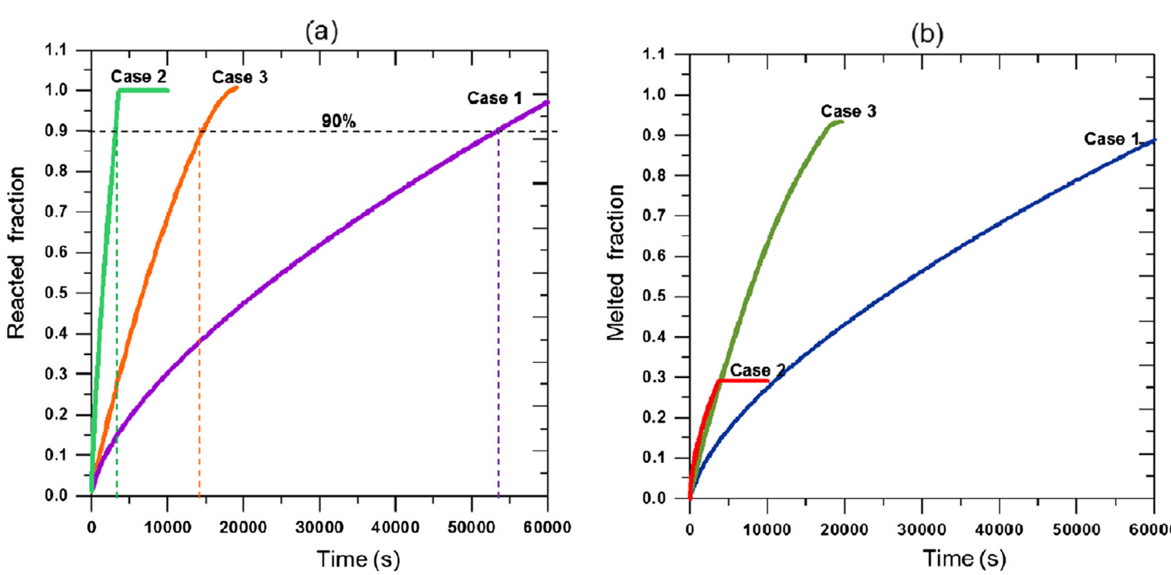

Fig. 9. Comparison of (a) the average reacted fraction and (b) the average liquid fraction for Case 1, 2 and 3.

Table 3
Thermal and physical properties of used HTFs [32].

Name of HTF	Melting point (°C)	Viscosity (Pa s)	Thermal conductivity (W m^{-1} K^{-1})	Heat capacity (kj kg^{-1} K^{-1})	Cost (\$/kg)	Corrosion		
						Rate (μm/year)	Alloy	Temperature (°C)
Air	–	0.00003	0.06	1.12	0	1.7–3.5	Fe-Al (5.8–16.2 *w*%)-Cr(1.9–9.7 *w*%)	1100
Water/steam	0	0.00133	0.08	2.42	0	1.7–3.5	In600	300
Thermal oil	−20	0.014	0.136	2.25	0.3	N/A	–	–
Molten-salt	95	0.007	0.654	1.44	0.62–0.81	8–12	316L	465

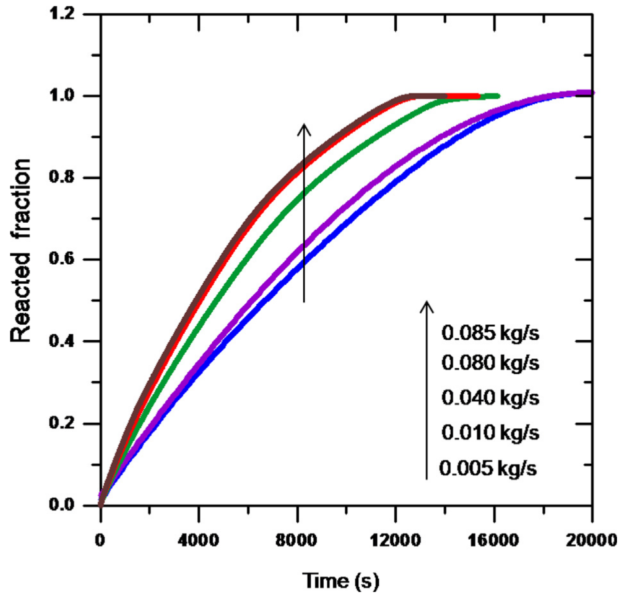

Fig. 11. Effect of HTF mass flow rate on average reacted fraction (for Case 3).

used in this simulation except the HTF mass flow rate was varied from 0.005 kg/s to 0.085 kg/s.

Effects of HTF mass flow rate through the U-type pipe are demonstrated in Fig. 11. Increasing the mass flow rate of HTF enhances the forced convection on the pipe wall, and thus improves heat transfer more efficiently. As indicated in Fig. 11, the time required for 90% of hydrogen storage capacity is about 14,685 s, 13,965 s, 10,695 s and 10,575 s for 0.005, 0.01, 0.04 and 0.08 kg/s, respectively. Thus, the charging time was reduced by 4%, 27% and 28% showing an improvement over the case with HTF flow rate equal to 0.005 kg/s.

From Fig. 11, it is found that increasing the mass flow rate from 0.005 to 0.010 kg/s reduced the filling time by 4% (for 90% hydrogen saturation), whereas when the mass flow rate was increased from 0.08 to 0.085 kg/s, there was no change in filling time. Also, it is observed that the absorption process can happen faster with increasing the mass flow rate in the pipe up to a particular value (0.08 kg/s). Thus, if the HTF pipe with more efficient heat transfer is desirable, increasing the mass flow rate more than this value have no significant effect. This can be explained by the fact that the heat removal in this system is limited by the thermal conductivity of the MH and PCM materials. Thus, it is important to maintain the mass flow rate of the HTF to this limit as there are no additional benefits by increasing the mass flow rate.

5. Conclusions

In the present study, a numerical study is carried out to investigate the benefits of employing a heat transfer fluid pipe in a Metal hydride tank equipped with a heat exchanger Phase Change Material. Three configurations are considered and the advantages and the limitations of each approach are evaluated with respect to their heat transfer performance and filling time of hydrogen vis-a-vis heat storage capacity. Through the obtained results, the following conclusions can be made:

1. To effectively integrate a thermal energy storage unit in a metal hydride-hydrogen storage tank, it is necessary to employ a HTF pipe which has a direct and significant influence in increasing both the heat transfer rate and the reaction rate. The computational results indicate a thermal coupling between the MH bed, the PCM medium and the HTF pipe.
2. It is found that the case where a HTF pipe extracts 70% of the heat of reaction and delivers it to environment as waste heat can reduce the hydrogen filling time by 94% compared to a case without heat pipe. However, the case where a HTF pipe extracts all the heat of reaction from MH bed and store it in a PCM, decreases the hydrogen filling time by 72% compared to a case without heat pipe.
3. The selection of a suitable HTF for the MH-PCM tank is important especially in a MH-PCM system in which heat transfer rate is a function of HTF type. Selection of an appropriate HTF can be one of the most effective design parameters to achieve a desired filling time.
4. Molten-salt can be an appropriate HTF that can be integrated in the MH-PCM tank for the best storage performance. However, the corrosion issues need to be resolved completely before its application is commercialised.
5. The mass flow rate is an important parameter affecting the hydrogen storage performance. The HTF circulated with a higher mass flow rate has a favorable performance of the MH-PCM tank. But, there is a limit beyond which increasing the mass flow rate does not lead to any significant improvements in hydrogen filling time. Thus, it is important to maintain the HTF flow rate above this limit.

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